

Vishwas Khattar

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Summary

Computer Engineering student at TIET (CGPA: 8.64) with strong DSA foundations (A-, 250+ LeetCode problems) and hands-on experience building backend and concurrent systems in C++, Node.js, and PostgreSQL. Built a multi-threaded ETL pipeline achieving 3.5x throughput gains and shipped production backend features during a live internship. Additional exposure to AI/ML applications and agentic workflows. Seeking Software Development Engineer opportunities.

Education

Thapar Institute of Engineering and Technology

B.E. Computer Engineering

2027

CGPA: 8.64

Modern Delhi Public School

PCM, CBSE

2023

Percentage: 87.8%

Technical Skills

- **Languages:** C, C++, Python, JavaScript, HTML/CSS
- **Web Tech:** Node.js, Express.js, React.js, REST APIs, JWT Auth, EJS, PostgreSQL, SQL
- **Tools:** Git/GitHub, Postman, VS Code, Google Earth Engine, GitHub Actions, Docker
- **AI/ML & Libraries:** Librosa, NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch, EfficientNet, CNN, OCR, n8n

Experience

Undergraduate Researcher – Edge AI & Audio ML

Ongoing

Thapar Institute of Engineering and Technology

- Authoring a research paper benchmarking audio ML models (MobileNetV2, EfficientNet-B1) deployed on Raspberry Pi 4B in low-resource forest environments; evaluating inference latency, accuracy, and power trade-offs across TFLite quantization formats (float32, float16, int8).

Backend Developer Intern – NextRound AI

June 2025 – Aug 2025

- Built backend infrastructure for an AI-powered UPSC prep platform: RESTful APIs, JWT authentication, and PostgreSQL operations using Node.js.
- Implemented content management and user progress tracking modules; optimized queries and collaborated with front-end to deliver end-to-end functionality.

Projects

- **ConcurrentETL - Parallel Data Pipeline** – C++, libpq, PostgreSQL, Neon [Link](#)
Designed a high-throughput ETL pipeline to ingest and load large batches of records into PostgreSQL while avoiding single-threaded I/O bottlenecks. Built a multi-stage parallel ETL pipeline using a producer-transformer-loader architecture with thread-safe queues (std::mutex, std::condition_variable) and raw libpq for direct PostgreSQL schema design. Benchmarked across 5 thread counts and 3 batch sizes, identifying database I/O as the primary bottleneck; optimized batch sizing achieving 3.5× throughput improvement over single-threaded baseline.
- **Sahayak – AI-Powered School ERP** – Node.js, Express.js, PostgreSQL, Gemini API, n8n [Link](#)
Built and deployed a full-stack school management platform on Render with JWT-secured, role-based portals for teachers and admins backed by PostgreSQL. Designed agentic workflows using the Gemini API and n8n orchestration to automate content generation, quiz creation, and clerical tasks. Configured deployment via Render Blueprint (render.yaml).
- **FRA-Atlas** – Python, OCR, Google Earth Engine, GIS [Demo](#)
Civic-tech platform to digitise Forest Rights Act documents and verify tribal land claims via OCR (text in Indian languages) and geospatial analysis. Processed 85+ legal documents across Chhattisgarh and Jharkhand (~91% OCR accuracy); mapped claims against 17 government welfare schemes using Google Earth Engine spatial eligibility workflows.
- **Edge AI Bird Species Classification** – Raspberry Pi 4B, TFLite, MobileNetV2, EfficientNet-B1, Python [Link](#)
Built a real-time edge AI system for bird species recognition using audio classification. Designed a four-stage inference pipeline with frequency-domain filtering, lightweight CNN-based detection, feature-extraction using librosa and multi-label classification. Achieved F1 score of 0.79 across 150+ species; reduced unnecessary inference by ~60% via audio energy gating. Benchmarked TFLite quantization formats (float32, float16, int8) for on-device deployment.

Certifications

• DSA with C++ (2024) - Coding Ninjas • Python Bootcamp - Udemy (2024) • Deep Learning - Coursera (2025)

Positions of Responsibility

Joint Secretary, Thapar MUN Society

May 2025 – May 2026

Managed logistics and delegate coordination for a 200+ member society under tight deadlines.

Executive Member, Urja Sports Fest

Jan 2024 – Feb 2024

Organised cultural and sports events for 300+ participants across a cross-functional team.